

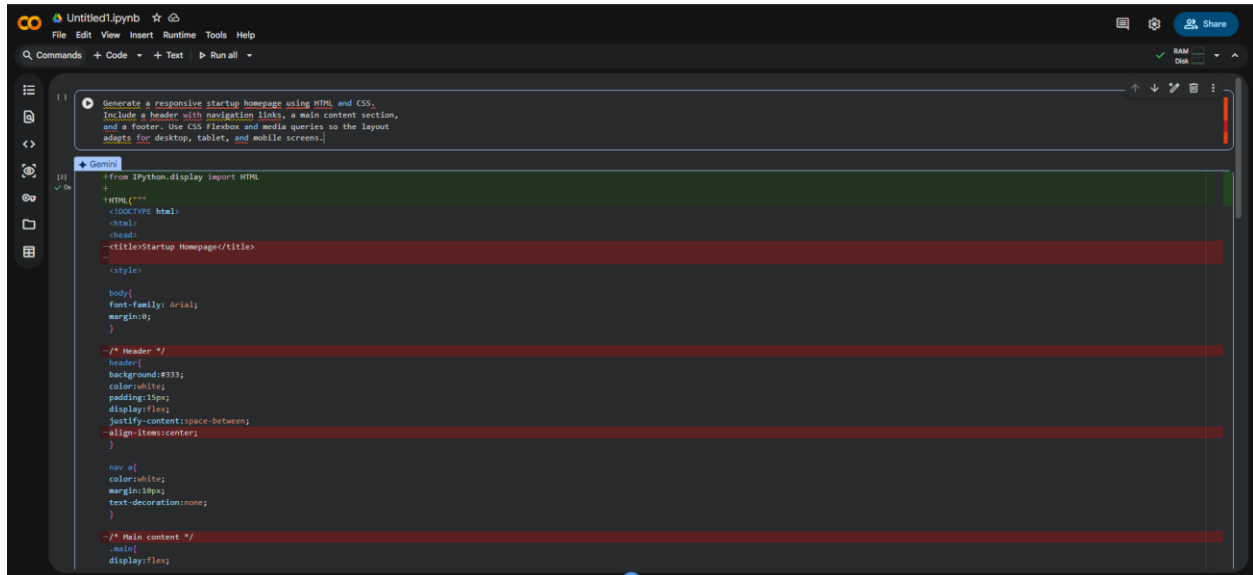
LAB ASSIGNMENT-14.4

2303A51602

Task 1: Responsive Homepage Layout for a Startup Website

Scenario:

You are designing a basic homepage for a startup company that needs to look good on desktops, tablets, and mobile devices.



```
Generate a responsive startup homepage using HTML and CSS.
Include a header with navigation links, a main content section,
and a footer. Use CSS flexbox and media queries so the layout
adapts for desktop, tablet, and mobile screens.

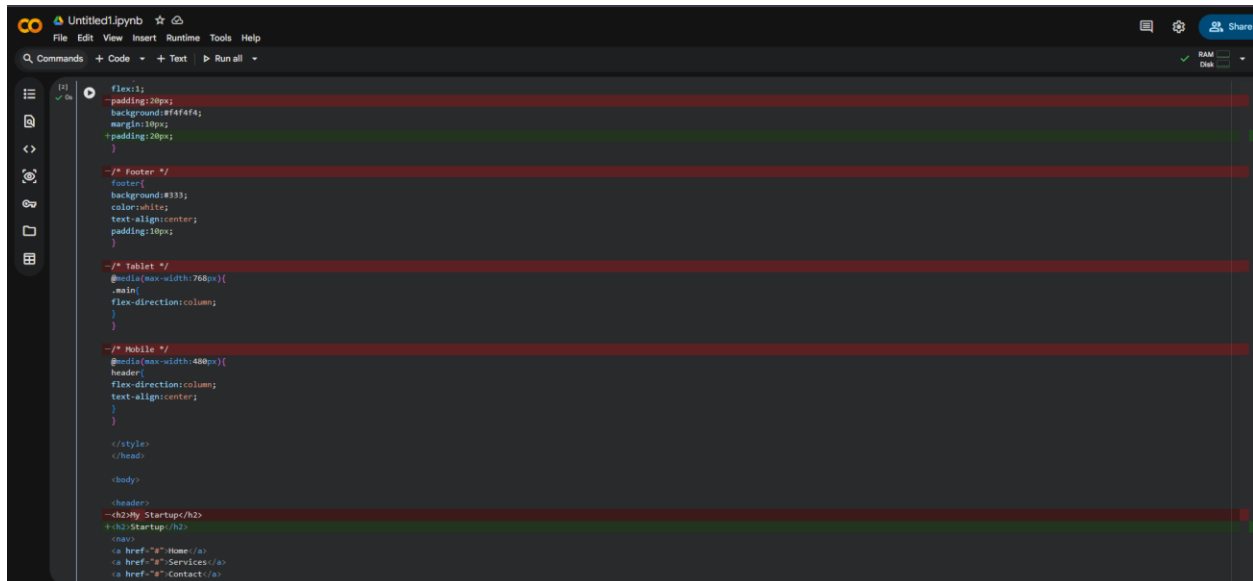
//from IPython.display import HTML
+
+HTML("""
<!DOCTYPE html>
<html>
<head>
<title>Startup Homepage</title>
<style>
body{
font-family: Arial;
margin:0;
}

/* Header */
header{
background:#333;
color:white;
padding:10px;
display:flex;
justify-content:space-between;
align-items:center;
}

nav ul
color:white;
margin:0px;
text-decoration:none;
}

/* Main content */
.main{
display:flex;

```



```
flex:1;
padding:20px;
background:#f4f4f4;
margin:10px;
padding:20px;

/* Footer */
footer{
background:#333;
color:white;
text-align:center;
padding:10px;
}

/* Tablet */
@media(max-width:768px){
.main{
flex-direction:column;
}
}

/* Mobile */
@media(max-width:480px){
header{
flex-direction:column;
text-align:center;
}
}

</style>
</head>

<body>

<header>
<h2>My Startup</h2>
<h2>Startup</h2>
<nav>
<a href="#">Home</a>
<a href="#">Services</a>
<a href="#">Contact</a>

```


Task 3: Contact Form with Client-Side Validation

Scenario:

You are developing a contact form for a company website to collect user queries safely and correctly

```
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File Edit View Insert Runtime Tools Help
Commands + Code + Text + Run all

1 Create a contact form with fields for name, email, and message.
   Use JavaScript to validate that fields are not empty and the email
   format is correct. Display error messages if input is invalid.

2 from Python.display import HTML

HTML("""
<html>
<body>

<h2>Contact Form</h2>

<form onsubmit="return validateForm()">

  Name:<br>
  <input type="text" id="name"><br>
  <span id="nameerror" style="color:red"></span><br>

  Email:<br>
  <input type="text" id="email"><br>
  <span id="emailerror" style="color:red"></span><br>

  Message:<br>
  <textarea id="message"></textarea><br>
  <span id="msgerror" style="color:red"></span><br><br>
  <button type="submit">Submit</button>
</form>

<p id="success" style="color:green"></p>

<script>

function validateForm(){

  let name=document.getElementById("name").value;
  let email=document.getElementById("email").value;
  let message=document.getElementById("message").value;


```

```
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Commands + Code + Text + Run all

3 let name=document.getElementById("name").value;
   let email=document.getElementById("email").value;
   let message=document.getElementById("message").value;

   let emailPattern=/^[a-zA-Z0-9]{2,3}$/;

   document.getElementById("nameerror").innerHTML="";
   document.getElementById("emailerror").innerHTML="";
   document.getElementById("msgerror").innerHTML="";

   if(name==""){
     document.getElementById("nameerror").innerHTML="Name required";
     return false;
   }

   if(!email.match(emailPattern)){
     document.getElementById("emailerror").innerHTML="Invalid email";
     return false;
   }

   if(message==""){
     document.getElementById("msgerror").innerHTML="Message required";
     return false;
   }

   document.getElementById("success").innerHTML="Form submitted successfully!";
   return false;
}

</script>
</body>
</html>
""")

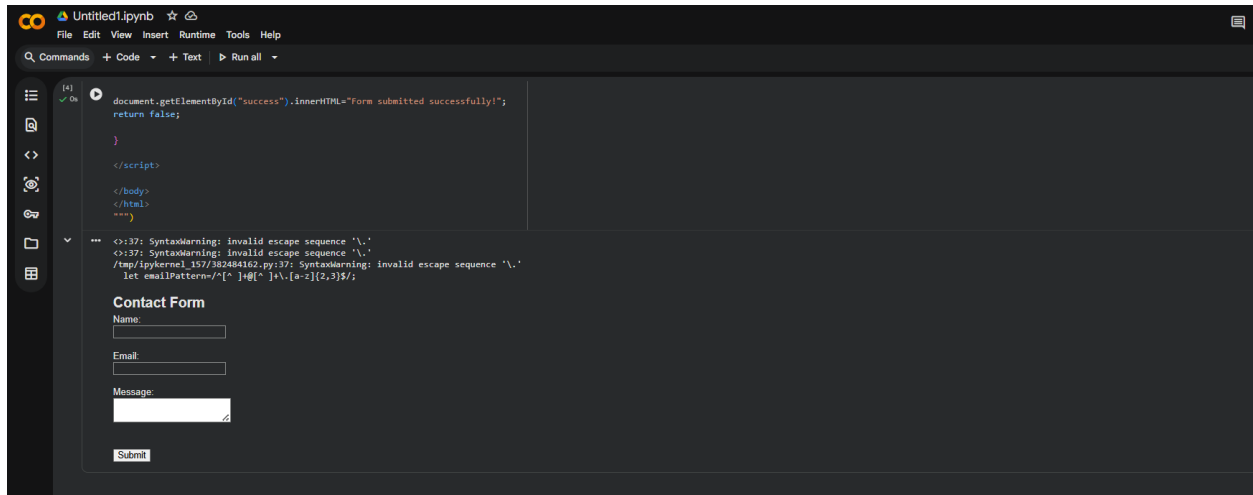
<!-- SyntaxWarning: invalid escape sequence '\.'
-->
<!-- SyntaxWarning: invalid escape sequence '\.'
-->
let emailPattern=/^[a-zA-Z0-9]{2,3}$/;

Contact Form
Name
```

Snipping Tool

Screenshot copied to clipboard
Automatically saved to screenshots folder.

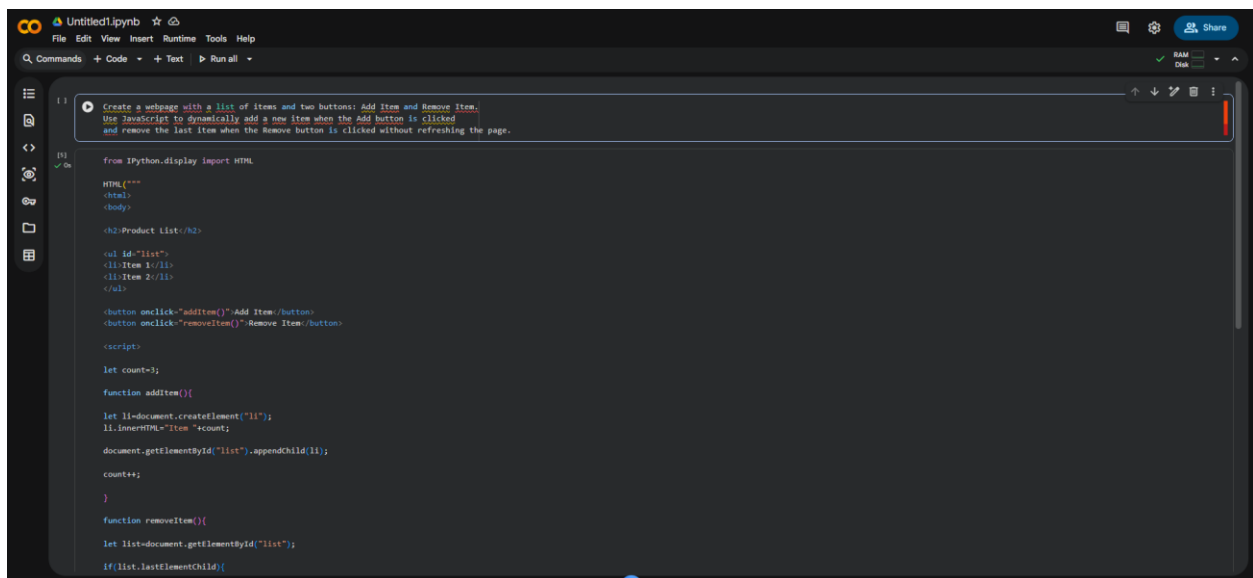
Markup and share

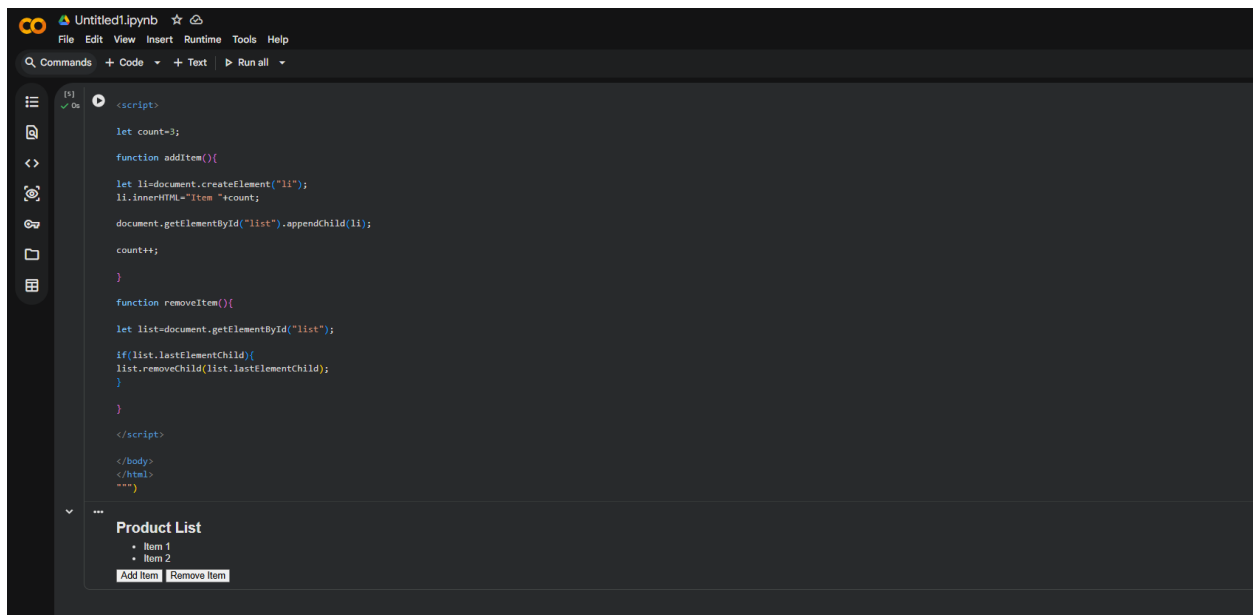


Task 4: Dynamic List Management for a Product Page

Scenario:

A product listing page needs to add or remove items dynamically without refreshing the page.





The screenshot shows a Jupyter Notebook interface with a dark theme. The top bar includes the Jupyter logo, the filename 'Untitled1.ipynb', and a star icon. Below the top bar is a menu bar with 'File', 'Edit', 'View', 'Insert', 'Runtime', 'Tools', and 'Help'. A search bar with the text 'Commands' is on the left. The main area contains a code cell with the following JavaScript code:

```
[1] ✓ On
<script>

let count=3;

function addItem(){

let li=document.createElement("li");
li.innerHTML="Item "+count;

document.getElementById("list").appendChild(li);

count++;

}

function removeItem(){

let list=document.getElementById("list");

if(list.lastElementChild){
list.removeChild(list.lastElementChild);
}

}

</script>

</body>
</html>
"""
```

Below the code cell, the rendered output is displayed:

Product List

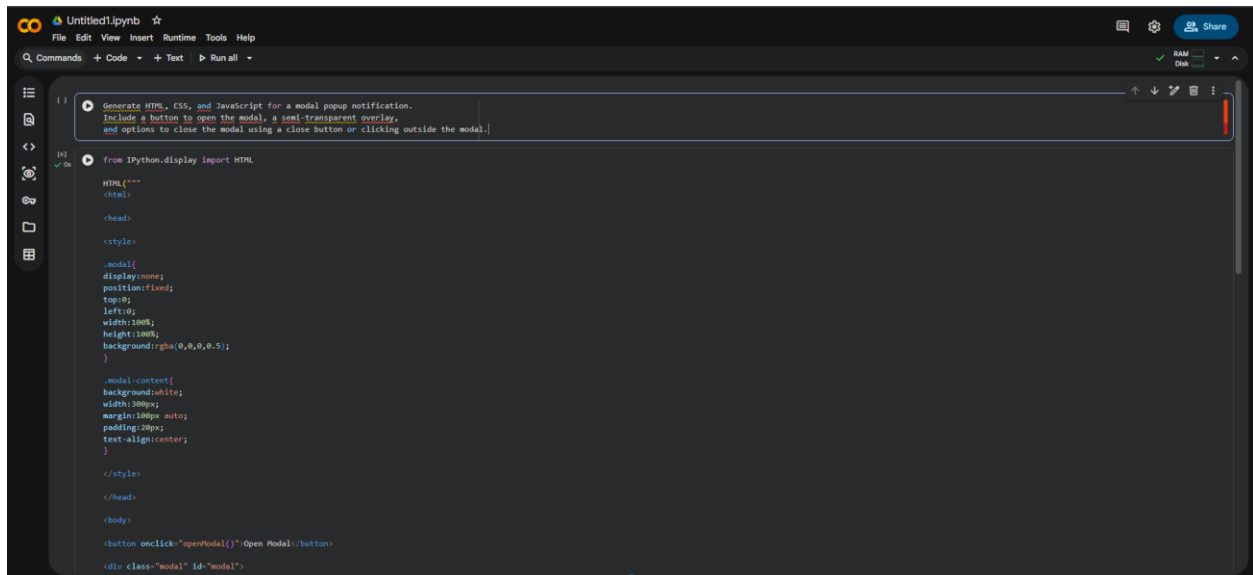
- Item 1
- Item 2

At the bottom of the rendered output, there are two buttons: 'Add Item' and 'Remove Item'.

Task 5: Modal Popup for User Notifications

Scenario:

You are building a user notification popup for announcements or alerts on a website.



The screenshot shows a Jupyter Notebook interface with a dark theme. The top bar includes the Jupyter logo, the filename 'Untitled1.ipynb', and a star icon. Below the top bar is a menu bar with 'File', 'Edit', 'View', 'Insert', 'Runtime', 'Tools', and 'Help'. A search bar with the text 'Commands' is on the left. The main area contains two code cells. The first cell contains a prompt:

```
[1] Generate HTML, CSS, and Javascript for a modal popup notification.
Include a button to open the modal, a semi-transparent overlay,
and options to close the modal using a close button or clicking outside the modal.
```

The second cell contains the following code:

```
[4] ✓ On
from IPython.display import HTML

HTML("""
<html>

<head>

<style>

.modal{
display:none;
position:fixed;
top:0;
left:0;
width:100%;
height:100%;
background:rgba(0,0,0,0.5);
}

.modal-content{
background:white;
width:300px;
margin:auto;
padding:20px;
text-align:center;
}

</style>

</head>

<body>

<button onclick="openModal()">Open Modal</button>

<div class="modal" id="modal">
```

Untitled1.ipynb

File Edit View Insert Runtime Tools Help

Commands + Code + Text ▶ Run all

RAM
Disk

Open Modal

```
<div class="modal-content">
  <h2 Notification /h2>
  <p>Welcome to our website!</p>
  <button onclick="closeModal()">Close</button>
</div>
</div>
<script>
function openModal(){
  document.getElementById("modal").style.display="block";
}

function closeModal(){
  document.getElementById("modal").style.display="none";
}

window.onclick=function(event){
  let modal=document.getElementById("modal");

  if(event.target==modal){
    modal.style.display="none";
  }
}
</script>
</body>
</html>
<!--
```